LEC02 - Structural Induction

Principle of Mathematical Induction for $\mathbb{N}$

Let $P : \mathbb{N} \to \mathbb{B} = \{T, \perp\}$ be a predicate. The predicate P holds over $\mathbb{N}$ iff the following is true:

1. The base case $P(0)$ is true
2. The inductive step: If $P(k-1)$ is true, then $P(k)$ is true, is true for all $k \in \mathbb{N} \setminus \{0\}$

Principle of Induction for Σ^*

Let $P : \Sigma^* \to \mathbb{B} = \{T, \perp\}$ be a predicate. The predicate P holds over Σ^* iff the following are true:

1. The base case $P(\varepsilon)$ is true
2. The inductive step: If $P(u)$ is true and $a \in \Sigma$, then $P(au)$ is true, is true for all $a \in \Sigma$ and $u \in \Sigma^* \setminus \{\varepsilon\}$

Principle of Mathematical Induction for general Inductively Defined Structures

Any IDS X provides a PMI for X

Structural Induction

Let X be the inclusion-minimal set satisfying:

1. Base clause ($X_0 \subseteq X$)
2. Inductive clause: For all $i = 1, 2, \cdots, m$, there are $(p_i, 1)$-ary relations R_i such that

$$x \in X^{p_i} \Rightarrow y_i := R_i(x) \in X$$

where $x = (x_1, x_2, x_3, \cdots, x_{p_i})$

Let X be an inductively-defined structure. Let $P : X \to \mathbb{B} = \{T, \perp\}$ be a predicate.

The predicate P holds over X iff the following are true:

1. Base case $P(x)$ is true for all $x \in X_0$
2. Inductive step: If $P(x_1), P(x_2), \cdots, P(x_{p_i})$ implies $P(y_i)$ is true, for all $i = 1, \cdots, m$

Example

Pal, set of all palindromes over $\{0,1\}$, $rev(w) = w$

Defined as the inclusion-minimal set such that

1. $\varepsilon \in Pal$, $0 \in Pal$, $1 \in Pal$
2. If $w \in Pal$, $0w0 \in Pal$ and $1w1 \in Pal$

$0 \to 000, 101$

$1 \to 010, 101$

$\varepsilon \to 11, 00$

Example

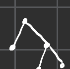
Consider the set $FBT = FBT[A]$ of all full binary trees over A, defined as the inclusion minimal set such that

1. The single node tree $tree(a)$, is in FBT for each $a \in A$
2. If $\ell, r \in FBT$, and $a \in A$, then $tree(a; \ell, r) \in FBT$

Example

The height $h(t)$ of a full binary tree t is recursively given by

$$h(t) := \begin{cases} 0 & \text{if } t = tree(a) \\ 1 + \max(h(\ell), h(r)) & \text{if } t = tree(\ell, r) \end{cases}$$



The size $n(t)$ of a full binary tree t is recursively given by

$$n(t) := \begin{cases} 1 & \text{if } t = tree(a) \\ 1 + n(\ell) + n(r) & \text{if } t = tree(\ell, r) \end{cases}$$

Theorem

For any FBT t, we have

$$n(t) \le 2^{h(t)+1} - 1$$

Proof by structural induction: Let $P(t): n(t) \le 2^{h(t)+1} - 1$

Base case: $t = tree(a)$

$P(t): 1 \le 2^{0+1} - 1$

$: 1 \le 1$, which is true

Inductive Step: Suppose $t = tree(a: \ell, r)$ and $P(\ell)$ and $P(r)$ are true

$$\begin{aligned} n(t) &= 1 + n(\ell) + n(r), \text{ where } n(\ell) \le 2^{h(\ell)+1} - 1,\ n(r) \le 2^{h(r)+1} - 1 \\ &\le 1 + 2^{h(\ell)+1} - 1 + 2^{h(r)+1} - 1 \\ &\le 2(2^{\max(h(\ell),h(r))+1} + 2^{\max(h(\ell),h(r))+1}) - 1 \\ &\le 2(2^{\max(h(\ell),h(r)+1)}) - 1 \\ &\le 2^{h(t)+1} - 1, \text{ as required} \end{aligned}$$

Conclusion:

By the PMI over FBT P is true over FBT

Noetherian Induction

- Generalization of strong induction

Over $\mathbb{N}$

Let $P: \mathbb{N} \to \mathbb{B} = \{T, \bot\}$ be a predicate, the predicate P holds over $\mathbb{N}$ iff

1. The base case $P(0)$ holds
2. The inductive step: If $P(j)$ is true for all $j < k$, then $P(k)$ is true, is true

Over Σ^*

Let $P: \Sigma^* \to \mathbb{B} = \{T, \bot\}$ be a predicate, the predicate P holds over Σ^* iff the following are true:

1. The base case $P(\varepsilon)$ is true
2. The inductive step: If $P(u)$ is true for all $u \prec v$, then $P(v)$ is true, is true

Generalization

Quasi-order: A relation $\preceq$ on A is called a quasi-order if it is reflexive and transitive

Reflexive: $\forall x \in A, x \preceq x$

Transitive: $x \preceq y$ and $y \preceq z \Rightarrow x \preceq z$

Well-founded quasi-order: A quasi-order $\preceq$ on A is called a well-founded quasi-order iff there are no strictly decreasing sequences

$$x_0 \succ x_1 \succ x_2 \succ \cdots$$

in A

$(\mathbb{N}, \le)$ is well-founded

$(\mathbb{N}_{>1}, |)$ is well-founded

divisibility

- Always terminate at prime numbers
- ex. $99 \succ 9 \succ 3$

$(\mathbb{Z}, |)$

- Not well founded because $1 \succ -1 \succ 1 \succ \cdots$

Noetherian Induction over a well-founded quasi-order

Let $P: X \to \mathbb{B} = \{T, \bot\}$ be a predicate. The predicate P holds over X iff the following are true:

1. The base case $P(x)$ is true for all minimal elements $x \in X$
2. The inductive step: If $P(x)$ is true for all $x \prec y$, then $P(y)$ is true, is true

Ackermann's (Partial) Function

Consider the (partial) function $A: \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ given by

$$A(x,y) := \begin{cases} y+1 & \text{if } x=0 \\ A(x-1,1) & \text{if } x \neq 0 \text{ and } y=0 \\ A(x-1, A(x,y-1)) & \text{otherwise} \end{cases}$$

Claim: The partial function A is a total function

Proof:

Base Case: $A(0,0) = 1$, which is well defined

Inductive Step: Assume $A(x',y')$ is well defined for $(x',y') \prec (x,y)$

Case 1: $x=0$ then $A(x,y) = y+1 \in \mathbb{N}$, so well defined

Case 2: $x \neq 0$ and $y=0$

Then $A(x,y) = A(x-1,0) \prec (x,0) \prec (x,y)$ is well defined by the inductive hypothesis

Case 3: $x \neq 0$ and $y \neq 0$

Then $A(x,y) = A(x-1, \overbrace{A(x,y-1)}^{\text{well defined by IH}}) = A(x-1,t) \prec (x,y)$ is well defined

$\square$